**Empirical Strategy:**

Data Structure:

Our analysis will be structured at the parent-by-child-by-year level. We denote a child *i*, a parent *j*, and a survey year *t*. The data are originally structured in a long format, with one observation for each person (parents and children) and each year, as shown below:

| id | father_id | mother_id | birth_year | Sex | syear | Outcome |
|---|---|---|---|---|---|---|
| 1 | 4 | 5 | 1999 | M | 2000 | |
| 1 | 4 | 5 | 1999 | M | 2001 | |
| 2 | 4 | 5 | 2001 | F | 2001 | |
| 4 | | | 1979 | M | 2000 | 9 |
| 4 | | | 1979 | M | 2001 | 10 |
| 5 | | | 1977 | F | 2000 | 8 |
| 5 | | | 1977 | F | 2001 | 7 |

We will merge child records to mother records and to father records, to create the following data structure:

| id | Parent_id | birth_year | Sex | Parent_Sex | syear | Outcome |
|---|---|---|---|---|---|---|
| 1 | 4 | 1999 | M | M | 2000 | 9 |

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | 4 | 1999 | M | M | 2001 | 10 |
| 2 | 4 | 2001 | F | M | 2001 | 10 |
| 1 | 5 | 1999 | M | F | 2000 | 8 |
| 1 | 5 | 1999 | M | F | 2001 | 7 |
| 2 | 5 | 2001 | F | F | 2001 | 7 |

Thus, a child with two parents will appear twice in each survey year. A parent with three children will appear three times in each survey year, in years after the birth of the last child.

Key Independent Variables:

We will determine children's birthdates with the variables birthd, birthm, and birthy_v2. Surveys will be included if conducted after the child's birth date.

We will determine parent and child sex with the variable pla0047_v2

We will identify mother and father with the variables mother_id and father_id, linked to the variable id.

Last, we will construct a categorical variable, prior_kids, that identifies the sex of all children born prior to child *i* to parent *j*. Consider a mother with four children, the first of whom is a boy, the next two of whom are girls, and the last of whom is a boy. prior_kids would take the following values:

| Birth_year | Sex | Prior_kids |
|---|---|---|
| 1999 | M | |
| 2000 | F | M |
| 2001 | F | MF |
| 2002 | M | MFF |

Note that prior_kids is defined at the parent-child level. A child with half-siblings will have a different value for prior_kids for mother observations and father observations.

Empirical Strategy:

There are a few reasons that the sex of a parent's children may be endogenous to the parent's attitudes or beliefs. First, individuals may differ in the likelihood that a child will be a boy, due to hormonal differences and the likelihood of sex-selective abortion. If these factors are related to gendered attitudes, parents with more sexist views may be more or less likely to produce boys than are parents with less sexist views. Second, a parent's decision to have an additional child may be influenced by the sex of their existing children. A parent who wants to have a boy may not have an additional child if they already have a boy, but will do so if they only have girls. Likewise for a parent who wants to have a girl.

We plan to address these problems by examining only child births after the first birth, and by controlling for the sexes of existing children. Because the sex of existing children provides information about the likelihood of a parent producing boys, this strategy reduces the first source of bias. Controlling for the sex of existing children accounts for the second source of bias.

We will run the following regression for each outcome:

$$Y_{i,j,t} = B_1 * f_i + \omega_{i,j} + X_{j,t} + \varepsilon_{i,j,t} \quad (1)$$

Where $Y_{i,j,t}$ is the outcome variable, $f_i$ is an indicator of whether child *i* is female, $\omega_{i,j,t}$ is a set of dummy variables for all values of Prior_kids, and $X_{j,t}$ is a series of controls for parent characteristics. $B_1$ is our coefficient of interest.

We will cluster the regression at the parent-level.

Sample Selection:

We will restrict the sample to second and later births, and to survey years after the birth of a child. We will not impose sample restrictions based on the age, marital status, or other characteristics of parents.

**Definition of Outcome Variables:**

We will define outcome variables for each hypothesis 1-3 as follows:

Hypothesis 1: Support for Gender Equality (attitudes):

We will use the variable **Plh0395i05**: "To what extent do you personally agree with the following statements? - Schools and preschools should promote non-gender-typical activities that are not typical for children's gender." This variable is coded with values of 1-7, with 7 expressing the strongest agreement with the statement. Negative values indicate missing values. We will drop all negative values and otherwise not recode the variable.

A positive coefficient $B_1$ from regression (1) will be interpreted as support for this hypothesis.

Hypothesis 2: Left-leaning political preferences:

We will use the variable **Plh0333**: "Which party did you vote for in the last general election (Bundestagswahl) on September 26, 2021?" This question was asked three times: in 2014, 2018, and 2022. We will use the responses in 2014 and 2018 (since 2022 is out of the sample range).

We will drop all missing values, and recode responses into four categories: 1: Right, 2: Center-Right, 3: Center-Left, and 4: Left. Responses will be coded based on parties' positions on social issues.

We plan to define categories in the following way:

1: Right: NPD, AFD

2: Center-Right: CDU, CSU

3: Center-Left: SPD, FDP

4: Left: Gruene Buendnis90, Pirate Party, Die Lienke

A positive coefficient $B_1$ from regression (1) will be interpreted as support for this hypothesis.

Hypothesis 3: Gender equality in the household (behavior):

For this hypothesis, our sample will be limited to heterosexual couples living together. We will combine several time-use variables to examine this hypothesis. In particular, we will use:

a. pli0019_h: [de] Childcare hours, Saturday [harmonized]

b. Pli0022_h: childcare hours, sundays

c. Pli0044_h: Childcare hours, weekdays

d. pli0012_h: [de] Housework hours/Saturday

e. Pli0016_h: Housework hours/Sunday

f. Pli0043_h: Housework hours/Weekdays

We will calculate each parent's total time spent on childcare and chores by first dropping all missing values and then calculating the following:

Total_time = pli0019_h + pli_0022_h + pli0012_h +pli0016_h + 5*(pli0044_h + pli0043_h)

Our final dependent variable will be the following, where total_time_m represents total_time for the male partner, and total_time_f represents total_time for the female partner:

$$time_{share_m} = \frac{\text{total_time}_\text{m}}{\text{total_time}_\text{m} + \text{total_time}_\text{f}}$$

Hypothesis 4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

For this hypothesis, we will use the outcome variable defined in hypothesis 1. We will run the following regression:

$$Y_{i,j,t} = B_1 * f_i + B_2 * f_i * m_j + \omega_{i,j} + \omega_{i,j} * m_j + X_{j,t} + X_{j,t} * m_j + \varepsilon_{i,j,t} \qquad (2)$$

Where $m_j$ is an indicator for whether the parent is male.

A positive coefficient $B_2$ from regression (2) will be interpreted as support for this hypothesis.

**Definitions of Control Variables:**

We plan to control for demographic and family characteristics that may be correlated with preferences around family size and with political beliefs. The broad categories we wish control for are: education, religiosity, income, and German nationality. Specific variables include:

plh0258_h – Categorial religious affiliation

plg0267_v2 – Categorical Educational Attainment

plb0471_h – Gross Income prev year

bhbula -categorical federal state

pl/plj0175: Both Parents Born In Germany

pl/plj0014_v3: German citizenship (yes/no)